Run report: claim grid 2026-10-14T18:54Z (descriptive)

Run. `research/ic_m3_algebraic_vs_mitm_20261014/run_grid.sh`, i.e. `ic_algebraic_floor run` for $m = 3$ and $m = 4$, run concurrently. **Code.** Instrument and protocol at commit `76e88621e`; runner at `ef4434914`. **Host.** Cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, Linux 6.18.44, rustc 1.97.0. Isolation L0: shared, the two grids concurrent, one thread per solve. **Time.** 18:54–20:26 UTC. **Workload.** Seed 20261014. Pools of 2,048 ($m = 3$) and 1,024 ($m = 4$). 6 decomposable and 6 non-decomposable targets per algebraic (arm, cell). Per target: 120 s and 4 GiB. Early stop after 4 leading censored targets.

This report records observations only; the verdicts are in the study README and `REPORT.pdf`.

Solves by verdict

Verdicts:

- `yes`: decomposed, witness checked.
- `no`: refuted.
- `none`: no verdict.
- `timeout`: killed at budget + 15 s.

There were no crashes and 0 disagreements with ground truth.

m	arm	yes	no	none	timeout
3	f4-chain	36	36	0	0
3	f6-ic	36	36	0	0
3	s4-hybrid-f10	24	18	6	4
3	s4-hybrid-f12	23	18	7	4
3	sat	18	9	13	0
4	f4-chain	24	24	3 (unsupported)	0
4	f6-ic	24	24	3 (unsupported)	0
4	s22-hybrid-g8	12	6	10	0
4	sat	15	11	14	0

Early stops, skips and unsupported cells

$m = 3$.

- `sat` stopped at $n = 23$ and was skipped at 29 and 31.
- Both S_4 hybrids stopped at $n = 29$ and were skipped at 31.

$m = 4$.

- `s22-hybrid-g8` stopped at $n = 17$.
- `sat` stopped at $n = 19$.
- F4 and F6-IC were unsupported at $n = 23, 29, 31$ (64-variable cap).

The floor equals exhaustive enumeration on 12/12 targets on every cell with $n \leq 23$.

Recorded values

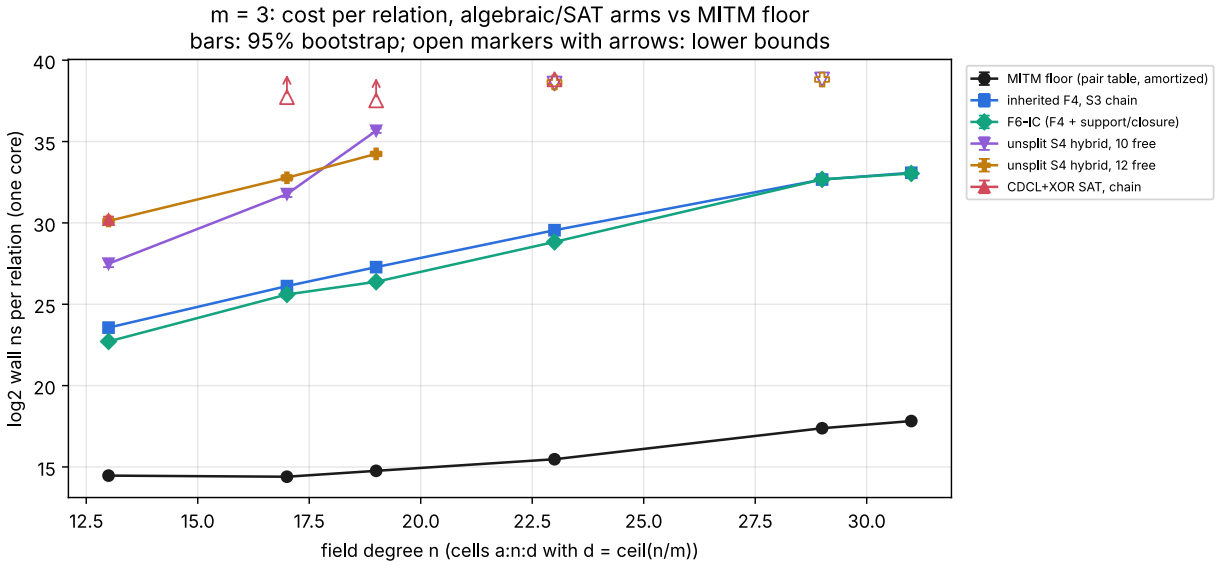


Figure 1: $m = 3$, $\log_2$ wall ns per relation (recorded; `ANALYSIS.json`).

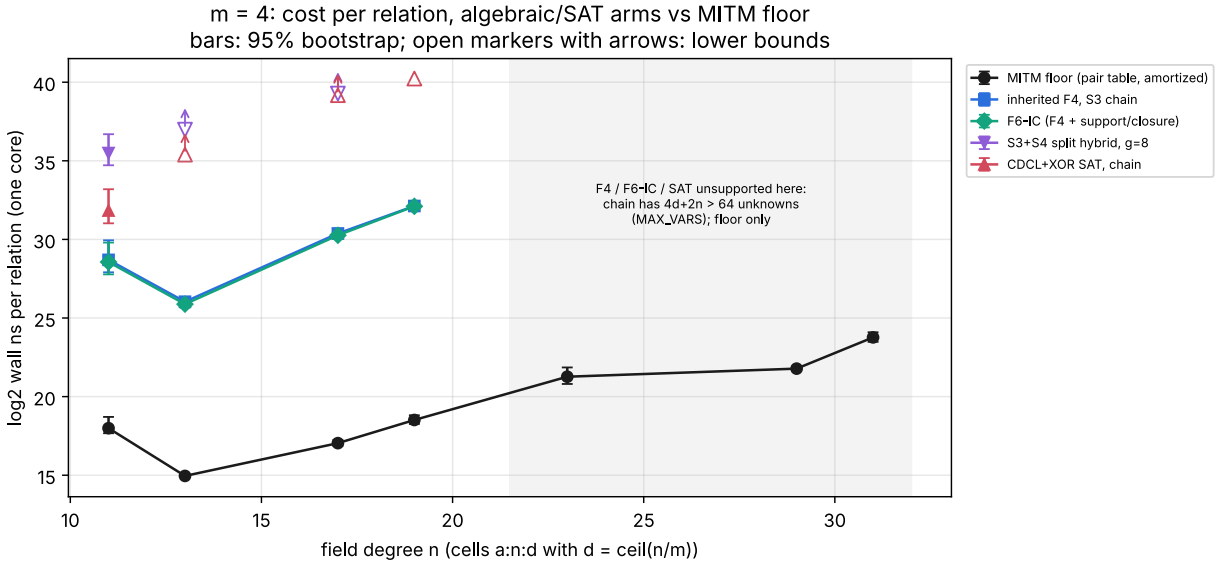


Figure 2: $m = 4$; shaded: only the floor ran.

Raw records:

- `runs/m3/` and `runs/m4/`: `truth-*.json` per cell, and `arm-*.jsonl` per (arm, cell).
- `runs-m3.log` and `runs-m4.log`.